Advanced Programming Practice · 21CSC203P

Unit 1 · Session 1 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 1 : SLO – 1 | Elements of Programming Languages

Activity (Solutions)

• 1) int x = 5 + 3; — SYNTAX: the grammatical structure of the statement: type keyword 'int', identifier 'x', assignment operator '=', expression '5 + 3' (two integer literals joined by the binary '+'), terminated by ';'. SEMANTICS: the meaning — evaluate 5 + 3 (value 8), allocate storage for an int variable x, and bind the value 8 to x.

• 2) Lexical tokens in 'int x = 5 + 3;' — keyword: int; identifier: x; operator: =, +; literals: 5, 3; separator/punctuator: ;. In a full program like public class Hello { public static void main(String[] a){ System.out.println("Hi"); } } the tokens are: keywords (public, class, static, void), identifiers (Hello, main, System, out, println), string literal ("Hi"), separators ({ } ( ) ; .).

• 3) Grammar rules demo: changing 'int x = 5 + 3;' to 'int x = (5 + 3) * 2;' follows the grammar rule expression = expression operator expression with parentheses for precedence; removing the semicolon breaks the grammar (compile error: ';' expected), showing that grammar rules define valid program structure.